

A GLOBAL SIGN FOR THE LITERAL ERDŐS–STRAUS RECEIVING DETERMINANT

THE CLANKERS

ABSTRACT. An explicit quartic receiving matrix used in a recent ES/Split-Zero construction has an additional determinantal divisor, distinct from its root discriminant. We prove that divisor is avoided by every positive integral ES witness at every prime $p \equiv 1 \pmod{12}$. Two exact polynomial certificates give uniform negative bounds and an original-coordinate inverse bound. Real and p -adic zeros do not contradict the result. No ES existence theorem is asserted.

For positive integers x, y, z with $4/p = 1/x + 1/y + 1/z$, retain $\ell = (p, x, y, z)$, $S = \sum \ell_i$, and the literal normalization

$$H(T) = -S^{-1} \prod_i (T - \ell_i) = AT^4 + T^3 + BT^2 + CT + D.$$

At simple roots choose and retain $\xi_i \in \mathbb{C}^\times$ with $\xi_i^2 H'(\ell_i) = 1$. The sign-odd receiving matrix $O \in M_4(\mathbb{C})$ in the supplied programme has columns

$$o_i = \xi_i(1, -id_i, Ad_i^2 + 2\ell_i d_i, i(7\ell_i^2 d_i - 13d_i^2))^T, \quad d_i = H'(\ell_i).$$

Put $V = \prod_{i < j} (\ell_j - \ell_i)$ and $\epsilon = A^2 V \prod_i \xi_i \in \{1, -1\}$. This retains root order and every square-root choice. The determinant is $\det O = 4\epsilon F/A^3 = -4\epsilon \mathfrak{N}/S^3$, where

$$\begin{aligned} F = & 20(3C - B^2) + A(87B^3 - 282BC - 51D) \\ & + A^2(-28B^4 + 22B^2C + 431C^2 + 204BD) \\ & + A^3(224B^2D - 168BC^2 - 856CD) - 448A^4D^2. \end{aligned}$$

Reduction of the four component polynomials modulo H proves this identity; the supplied standard-library verifier checks it universally. Set $\mathfrak{N}(\ell) = S^6 F$. This is an integer homogeneous polynomial of degree eight in integral ℓ , without any assumption that S is a p -unit.

Theorem 1. *For every prime $p \equiv 1 \pmod{12}$ and every positive integral ES witness,*

$$\boxed{\mathfrak{N} < -\frac{125873811}{262144} p^8 < 0.}$$

For $p \equiv 1 \pmod{24}$ the stronger constant is $c = 327448292668/47045881$. With either applicable constant,

$$\boxed{|\det O| > \frac{4cp^8}{S^3}, \quad \|O^{-1}\|_2 < \frac{360S^9}{cp^8}.}$$

Here $\|\cdot\|_2$ is the operator norm for the standard Hermitian norm on $\mathbb{C}^4$.

The complete original arithmetic domain. Four literal roots are distinct: $x = p$ would give $(3y - p)(3z - p) = p^2$ with incompatible residues modulo three; $x = y = a$ gives $4(2a - p)z = p(2a)$ and $2a - p \mid p^2$, whose three cases all fail integrality for $p \equiv 1 \pmod{4}$. At least one, but not all, denominators are divisible by p .

If two are, the third is the unique smallest denominator $a < p/2$. If only one is pk , let the other two be $a \leq b$, both units at p . Write $4k - 1 = pt$, $a = gr$, $b = gs$, $(r, s) = 1$. Then $k = drs$, $r + s = t\lambda$, $g = d\lambda$, and $t \equiv 3 \pmod{4}$. The contrary inequality $a \geq p/2$ gives $2dr(r - s) \geq -1$, hence $r = s = 1$ and $t \mid 2$, impossible. Thus every witness has $p/4 < a < p/2$.

Put $R = 4a - p$. The identity $(Ry - pa)(Rz - pa) = p^2a^2$ recovers the full original Type I/II source [1, §2], with $u \mid a^2$:

	gate	ordered denominator return
E	$R \mid 4u + 1$	$(a, (pa + a^2/u)/R, (pa + p^2u)/R)$
M	$R \mid a + u$	$(a, p(a + a^2/u)/R, p(a + u)/R)$

The original normalization is $(h, r, s) = (g^2/u, u/g, a/g)$, $g = (a, u)$. The ordered divisor inverses are $a^2/(Ry - pa)$ and $pa^2/(Ry - pa)$. Complementation exchanges only the two M tails and is retained.

An E state has $u \geq 2$: $u = 1$ would require $R \mid 5$. Write $(\cdot/R)$ for the Jacobi symbol and $(\cdot/p)$ for the Legendre symbol. Reciprocity primewise on a gives $(u/R) = (u/p)$, so its target $-1/4$ makes $(u/p) = -1$. When $p \equiv 1 \pmod{24}$, two and three are residues, and therefore $u \geq 5$. For M, write the two tails as $pb < pc$. Since neither denominator is p , $b \geq 2$ and $c - b \geq 1$ are original integer inequalities.

Exterior coefficient identity. Use real $u \geq 1$ and $w = p/R > 1$. The four normalized roots are

$$\left(1, \frac{w+1}{4w}, \frac{w+1}{4} + \frac{(w+1)^2}{16wu}, \frac{w+1}{4} + wu\right).$$

Define $P_E \in \mathbb{Z}[u, w]$ by the exact division

$$\begin{aligned} -64u^2P_E(u, w) &= \mathfrak{N}(16wu, 4u(w+1), \\ &\quad (w+1)(4wu+w+1), 4wu(w+1+4wu)). \end{aligned}$$

It has bidegree (12, 16). Each of the following has 221 positive integer coefficients, occupying the full 13×17 rectangle:

$$P_E(1+X, 1+Y), \quad (uP_{E,u} - 6P_E)(1+X, 1+Y), \quad (wP_{E,w} - 8P_E)(1+X, 1+Y).$$

The arrays are the finite certificate. Hence

$$Q_E(u, w) = \frac{P_E(u, w)}{2^{26}u^6w^8} = -\frac{\mathfrak{N}}{p^8}$$

is strictly increasing in both arguments. Exact boundary evaluation gives

$$Q_E(2, 1) = \frac{125873811}{262144}, \quad Q_E(5, 1) = \frac{1310003654463}{12800000} > \frac{327448292668}{47045881}.$$

This proves both required bounds in the exterior channel.

Middle coefficient identity. Let $L = 4bc - b - c$, $T = (b+c)L + bc$, $V = b^2c^2$. Then $a/p = bc/L$ and $-\mathfrak{N}/p^8 = \mathcal{P}(T/L, V/L)$, where

$$\begin{aligned} \mathcal{P}(s, v) &= 7168v^4 - (5568s^2 + 5728s - 5888)v^3 \\ &\quad + (320s^4 + 2744s^3 + 705s^2 - 3350s - 903)v^2 \\ &\quad + (-140s^5 - 443s^4 + 440s^3 + 390s^2 + 70s - 249)v \\ &\quad + 20s^6 - 7s^5 - 26s^4 - 7s^3 + 20s^2. \end{aligned}$$

Put

$$H_M(X, Y) = \left[L^6 \mathcal{P}(T/L, V/L) \right]_{b=2+X, c=3+X+Y},$$

$$L_M = 4X^2 + 4XY + 18X + 7Y + 19.$$

The polynomial $47045881H_M - 327448292668L_M^6$ has zero constant and 191 positive nonzero integer coefficients. Its X and Y coefficients are positive. Dividing by $47045881L_M^6$ proves the stronger bound for all $X, Y \geq 0$. Equality requires $b = 2, c = 3$, hence $a = 6p/19$, which is not integral at a prime $p \equiv 1 \pmod{12}$. The inequality in the theorem is strict.

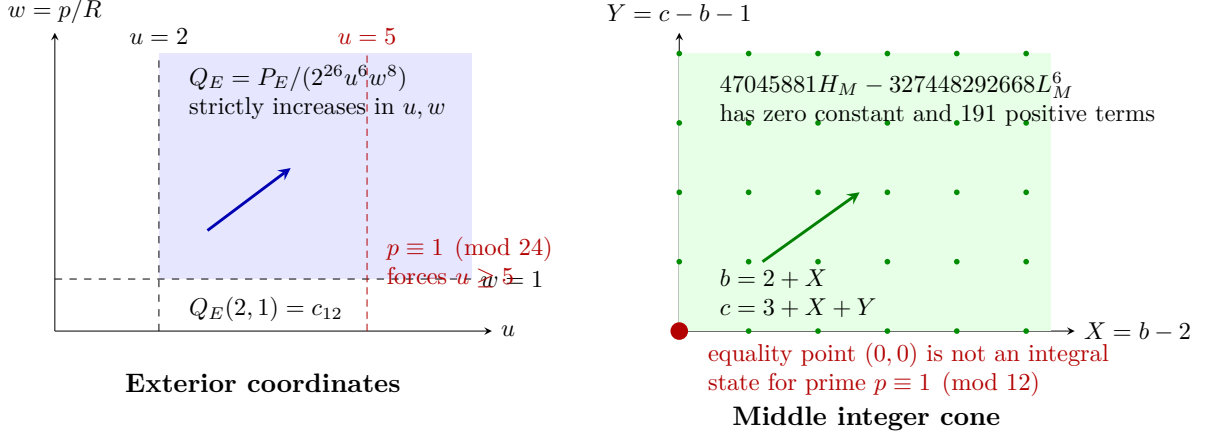


FIGURE 1. The two exact coordinate domains used in the global sign proof. The blue and green regions are certificate domains; actual Erdős–Straus witnesses are the arithmetic points satisfying the displayed gates and integrality conditions. The exterior lower bound is approached at the dashed boundary $w = 1$, which is excluded by $R < p$. The middle comparison is strict at every admissible lattice point because its only equality point would require $a = 6p/19$.

Why the finite certificate is a proof. The source norm is the literal F of [3, Track I, E1], not a fitted array. The main program expands every rational coefficient identity exactly, then checks all its coefficients. The separate implementation proves the exterior identity on the entire 17×17 unisolvent grid after clearing its known bidegree $(16, 16)$ denominator. It independently performs the two derivative and binomial-translation operations. For M, it checks

$$\mathfrak{N}(L, bc, bL, cL) = -L^2 H_M$$

on the full 25×17 grid in X, Y , covering bidegree $(24, 16)$, then independently expands the comparison with L_M^6 . Repeated one-variable polynomial uniqueness makes these universal identities. This is a computer-assisted coefficient lemma, not a finite-prime extrapolation. Original inverse and norm. Let C_H be the reduction matrix of the four original component polynomials, including their i factors, and $U_{nj} = \ell_j^n \xi_j$. Then $O = C_H U$. Given z , set $m = C_H^{-1} z$ using its adjugate. Lagrange interpolation returns

$$\begin{aligned} \beta_j = & \xi_j [(C + B\ell_j + \ell_j^2 + A\ell_j^3)m_0 + (B + \ell_j + A\ell_j^2)m_1 \\ & + (1 + A\ell_j)m_2 + Am_3]. \end{aligned}$$

For four distinct integer roots, $2/S \leq |d_j| \leq S^2$. The entry bounds by row are $\sqrt{S/2}$, S , $3S^2$ and $20S^3$. Every cofactor is at most $360S^6$; the adjugate Frobenius norm is at most $1440S^6$. Dividing by $4cp^8/S^3$ proves the inverse bound. Original positive metrics G, Q contribute the explicit factor $\|G^{1/2}\| \|Q^{-1/2}\|$.

Scope and negative controls. At $p = 1009$, the genuine E and M returns from $(a, u) = (253, 11)$ are $(253, 87032, 3818056)$ and $(253, 2042216, 88792)$. They violate two of the audit's three sufficient character conditions, but are covered here. This does not assert $p \nmid \mathfrak{N}$: their valuations are zero and two.

At $p = 1$, let $b = t$, $c = t + 1/10$, $a = bc/(4bc - b - c)$ for $3 \leq t \leq 4$. The literal roots stay positive and distinct. Exact elimination produces a degree-18 numerator $P(t)$ with all coefficients of $P'(3 + X)$ positive and opposite signs at $3053/1000$ and $1527/500$. Hence this real family has one receiver singularity, at $t \approx 3.05367681075890$. Its tail gap is not the original integer gap used above. The earlier local zeros likewise remain local, not integral, counterdomains.

Finally $p = 1009, a = 321, u = 9$ gives the unfiltered E triple $(321, 335338/275, 9486618/275)$. Its frame is nonsingular, but its E gate is not satisfied: $275 \nmid 37$. Its $S^6 F$ is rational, not an integer numerator. Thus the theorem excludes receiver singularity conditional on an existing

positive integral witness at $p \equiv 1 \pmod{12}$; it does not establish universal occupancy of the ES source. No priority determination, independent human review, or Lean build is claimed.

REFERENCES

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